MILITARY STANDARD

MIL-STD-1320-190 (NAVY)

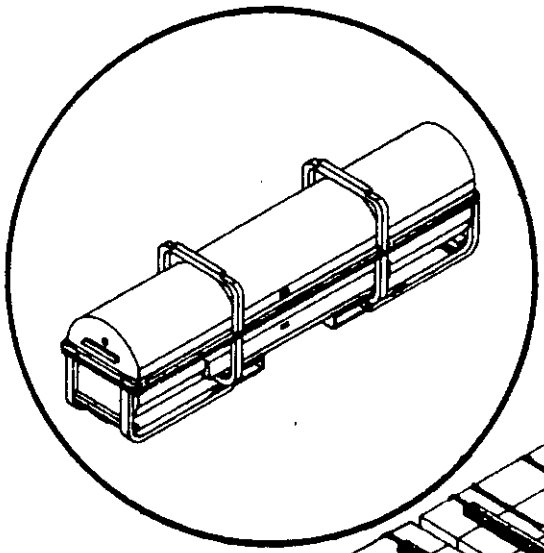
TRUCKLOADING

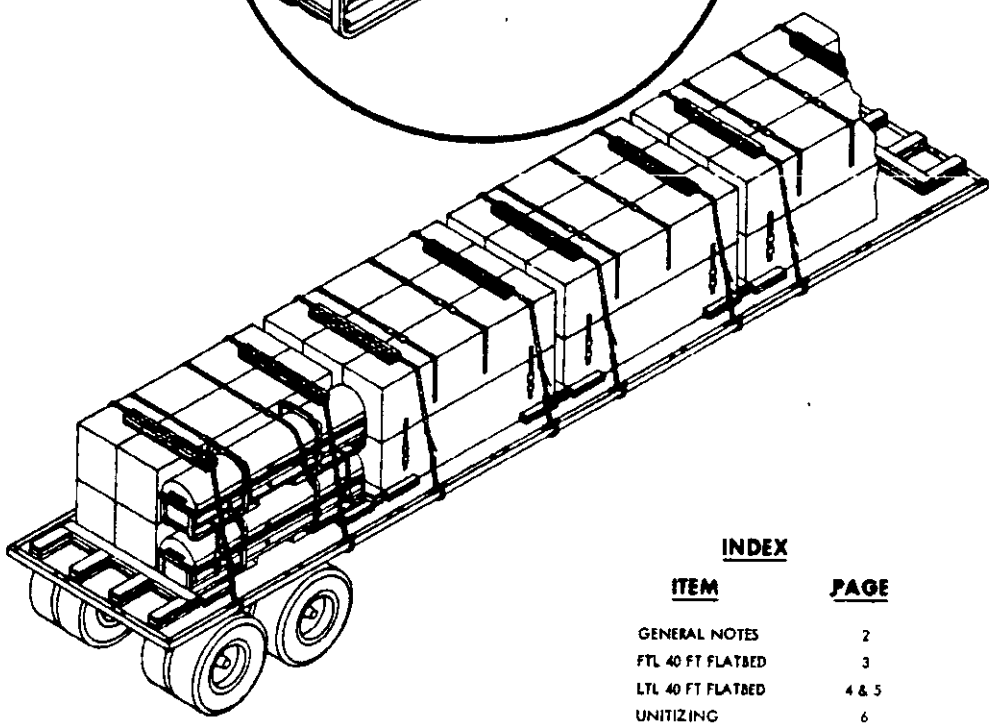
20 MARCH 1978

SUSTAINER SECTION A/B44G-1
HARPOON GUIDED MISSILE
IN CONTAINER MK 621 MOD 0

CONTAINER DATA

DIMENSIONS ..... 103 L X 23 5/8 W X 28 H
CUBE ..... 39.4 CU. FT.
GROSS WT (APPROX.)
W/BOATAIL .....1200 LBS.
W/O BOATAIL ..... 972 LBS.
DOT HAZARD CLASSIFICATION ..... EXPLOSIVES A
CONTAINER DWG. ..... DL 2645395





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NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENT ONLY

W.E. Gray NWEIC 3/6/78
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

SIGNATURE SEA SYSCOM, BY DIRECTION DATE

ORIGINATOR 2 MAR 1978
DATE

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

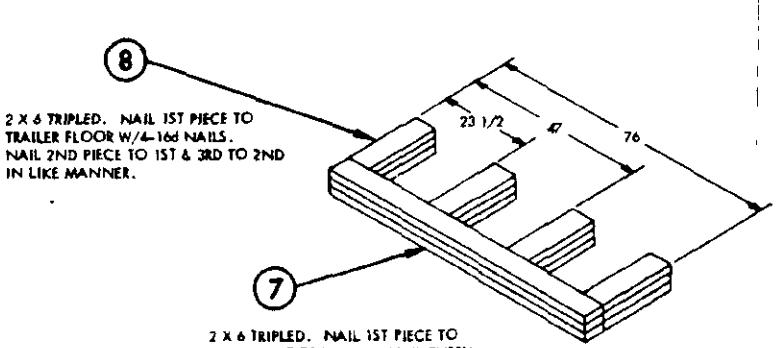
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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE HARPOON SUSTAINER SECTION A/B44G-1 WITH OR WITHOUT BOATAIL IN THE CONTAINER MK 621 MOD 0.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT. FLATBED TRAILERS WITH THE TRAILER'S TANDEM AXLES LOCATED IN THE "WESTERN POSITION" (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. WHEN CONTAINERS ARE TWO HIGH, THEY SHALL BE UNITIZED AS SHOWN ON PAGE 6.
4. CHAINS AND STEEL STRAP MAY BE USED INTERCHANGEABLY ON A 1 TO 1 BASIS.
5. STEEL STRAP USED FOR TIEDOWNS SHALL BE 2 X .050 AND SHALL BE APPLIED AS SPECIFIED IN THE BASIC TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY), AND AS SHOWN ON PAGE 5 OF THIS DOCUMENT.
6. CHAINS AND LOADBINDERS SHALL BE AS SPECIFIED IN THE BASIC TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY) AND APPLIED AS SHOWN IN THIS DOCUMENT AND THE BASIC DOCUMENT MIL-STD-1320 (NAVY).
7. PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY NAVWEPS OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS, AND LOADBINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOADBINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOADBINDER.
8. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
9. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO DUNNAGE IN AN ACCESSIBLE AREA AND ATTACH "EXPLOSIVES A" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
10. APPLICABLE MATERIAL SPECIFICATIONS:
    DUNNAGE LUMBER, MM-L-751
    NAILS, FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT
    STRAPPING, QQ-S-781, TYPE I, HEAVY DUTY, FINISH A, B OR C
11. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

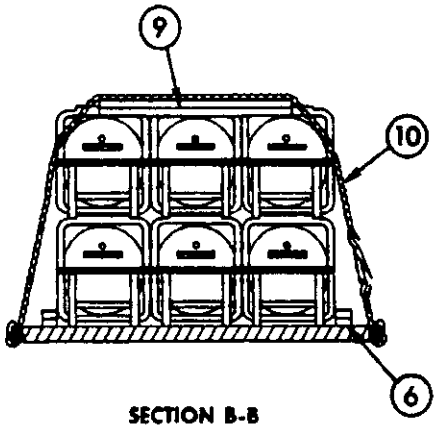
FTL 40 FT TRAILER (FLATBED)

1. CONTAINERS SHOULD BE STACKED TWO HIGH AND STRAPPED TOGETHER BEFORE LOADING ON TRAILER. FOR UNITIZING DETAILS, SEE PAGE 6.
2. STRAP TOP LAYER OF EACH STACK OF CONTAINERS TOGETHER WITH TWO 1 1/4" STRAPS (PIECE NO. 3) IN POSITION SHOWN. AFTER TENSIONING STRAPS SEAL WITH TWO 1 1/4" SEALS (PIECE NO. 4). DOUBLE CRIMP SEALS.



DETAIL A

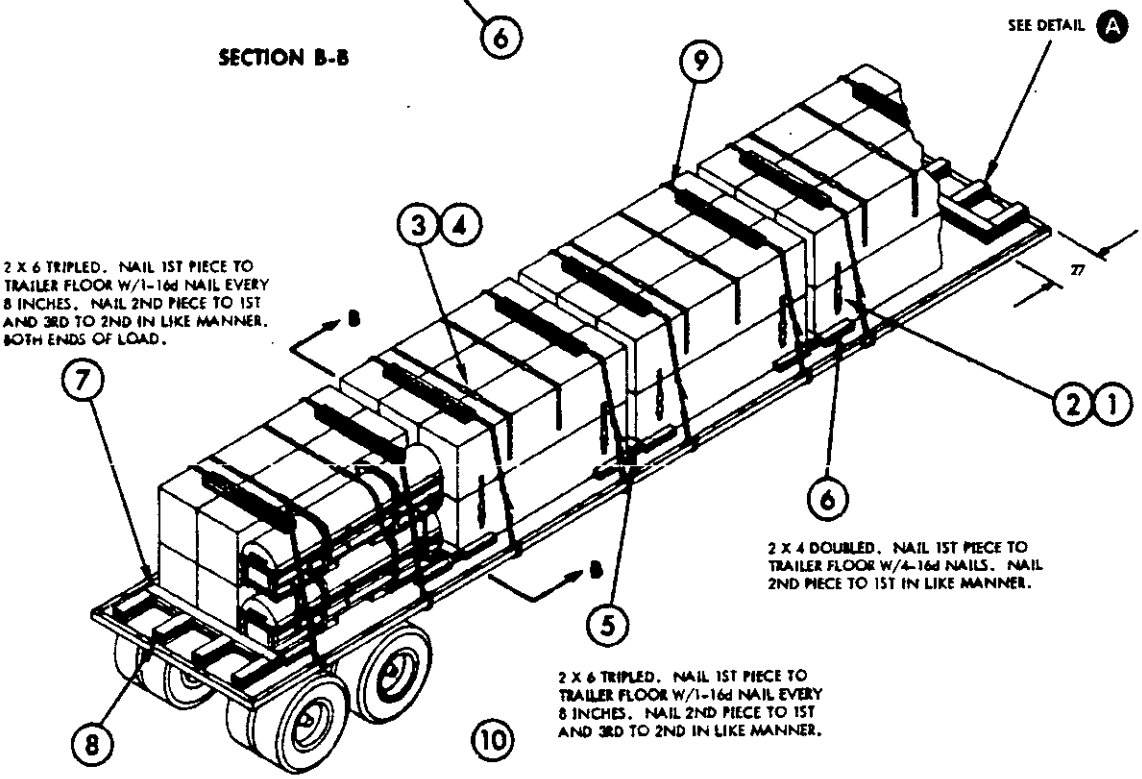
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SECTION B-B

NOTE

THIS FTL ILLUSTRATES THE USE OF CHAINS AND LOADBINDERS. 2 X .050 STRAPS ARE AN AUTHORIZED ALTERNATE. SEE GENERAL NOTE 4.



TRUCKLOAD DATA

NO. OF CONTAINERS.....24
WEIGHT (APPROX.) SUSTAINER
WITH BOATAIL .....24,480 LBS.
W/O BOATAIL.....23,328 LBS.

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
10	CHAIN & LOADBINDER	5/16 OR 3/8	8	-	-	-
9	PROTECTOR BOARD	2 X 6 X 54	16	DOUBLE LAMINATE	6	8d
8	BACKUP CLEAT	2 X 6 X 20	24	SEE FIELD NOTE		
7	END CROSSMEMBER	2 X 6 X 76	6	SEE FIELD NOTE		
6	SLEEPER	2 X 4 X 18	32	SEE FIELD NOTE		
5	SEPARATOR	2 X 6 X 76	9	SEE FIELD NOTE		
4	SEAL	FOR 1 1/4 STRAP	16	-	-	-
3	CROSS STRAP	1 1/4 X .035 X 18 FT.	8	-	-	-
2	SEAL	FOR 1 1/4 STRAP	48	-	-	-
1	UNITIZING STRAP	1 1/4 X .035 X 11 FT.	24	-	-	-

LIST OF MATERIALS & NAILING DATA

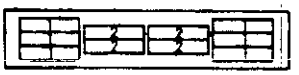
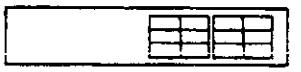
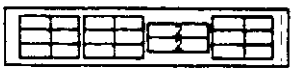
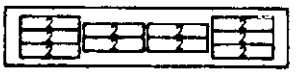
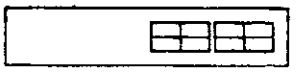
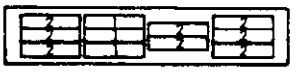
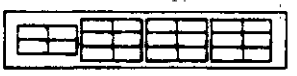
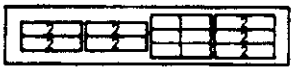
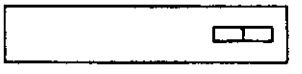
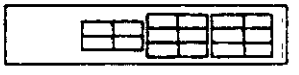
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LTL 40 FT TRAILER (FLATBED)

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF CONTAINERS TO BE SHIPPED FROM TABLE I AND POSITION CONTAINERS ON THE TRAILER ACCORDINGLY.
2. WHEN LOADING 23 CONTAINER, THE TOP LAYER OF THE THIRD STACK REQUIRES A FILLER ASSEMBLY (SEE DETAIL D, PAGE 6). STRAP THE ASSEMBLY TO THE LOWER CONTAINER WITH TWO 1 1/4 X .035 STEEL STRAPS. THE FILLER ASSEMBLY UPRIGHTS SHALL BE OPPOSITE THE ADJACENT CONTAINER ROLLRINGS.
3. THE LTL ON PAGE 5 IS FOR 16 CONTAINERS AND SHOWS THE CORRECT BLOCKING AND TIEDOWNS FOR THIS QUANTITY. WHEN LOADING OTHER QUANTITIES, BLOCK AND TIEDOWN USING THE PRINCIPLES SHOWN ON PAGES 3 AND 5.
4. THE LTL SHOWN ON PAGE 5 DOES NOT LIMIT LTL SHIPMENTS TO THIS QUANTITY. ITS PRESENTATION IS INTENDED TO EXPLAIN THE CORRECT METHODS OF LOADING LTL SHIPMENTS OF THIS CONFIGURATION.

TABLE I

LOAD PATTERN*

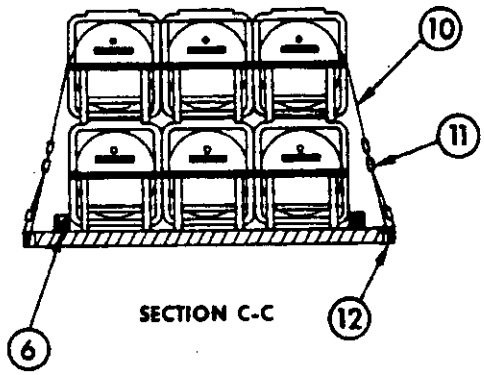
NO. OF CONTAINERS → FWD		NO. OF CONTAINERS → FWD		NO. OF CONTAINERS → FWD	
23		15		7	
22		14		6	
21		13		5	
20		12		4	
19		11		3	
18		10		2	
17		9		1	
16		8			

*LOAD PATTERN SHOWS FLOOR PLAN FOR TRAILER. NUMBER INDICATES THE NUMBER OF CONTAINERS IN A LAYER.

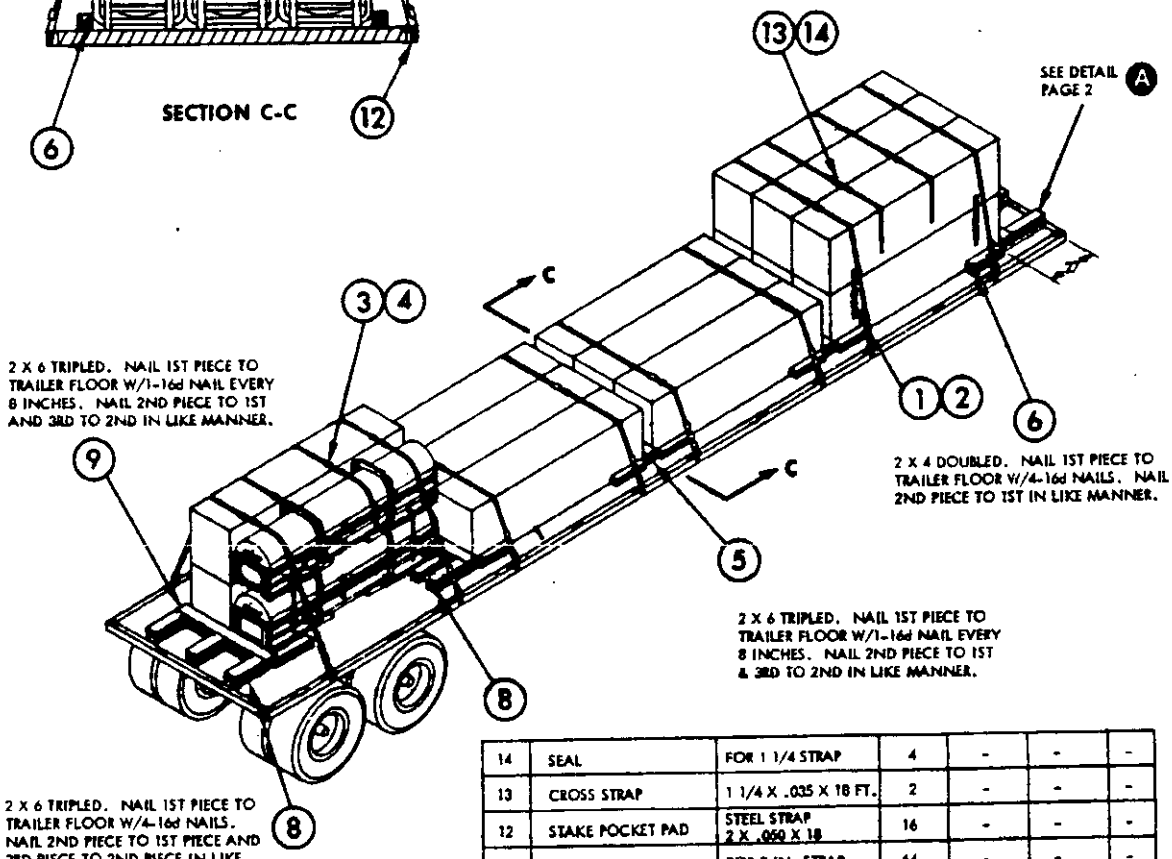
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NOTE

THIS LTL ILLUSTRATES THE USE OF 2 X .050 STRAPS AS TIEDOWNS. CHAINS ARE AN AUTHORIZED ALTERNATE. SEE GENERAL NOTE 4.



SECTION C-C



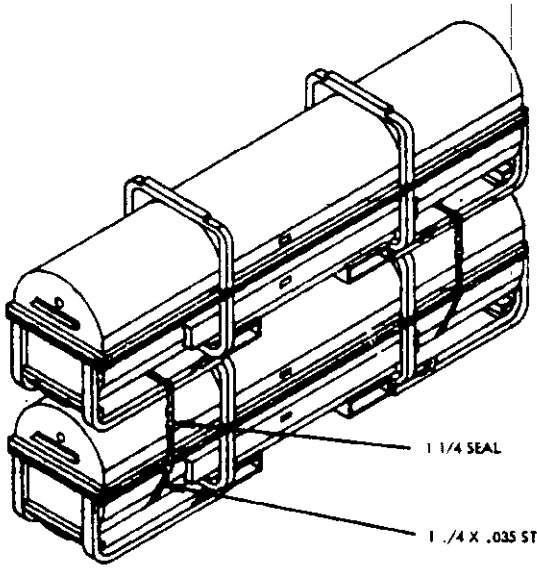
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
14	SEAL	FOR 1 1/4 STRAP	4	-	-	-
13	CROSS STRAP	1 1/4 X .035 X 18 FT.	2	-	-	-
12	STAKE POCKET PAD	STEEL STRAP 2 X .050 X 18	16	-	-	-
11	SEAL	FOR 2 IN. STRAP	64	-	-	-
10	STRAP	2 X .050 X 20 FT.	8	-	-	-
9	AFT CROSSMEMBER	2 X 6 X 53	3	SEE FIELD NOTE		
8	BACKUP CLEAT	2 X 6 X 20	27	SEE FIELD NOTE		
7	FWD CROSSMEMBER	2 X 6 X 76	3	SEE DETAIL A		
6	SLEEPER	2 X 4 X 18	32	SEE FIELD NOTE		
5	SEPARATOR	2 X 6 X 76	9	SEE FIELD NOTE		
4	SEAL	FOR 1 1/4 STRAP	4	-	-	-
3	CROSS STRAP	1 1/4 X .035 X 14 FT.	2	-	-	-
2	SEAL	FOR 1 1/4 STRAP	20	-	-	-
1	UNITIZING STRAP	1 1/4 X .035 X 11 FT.	10	-	-	-

LIST OF MATERIALS & NAILING DATA

TYPICAL LTL 40 FT TRAILER (FLATBED)

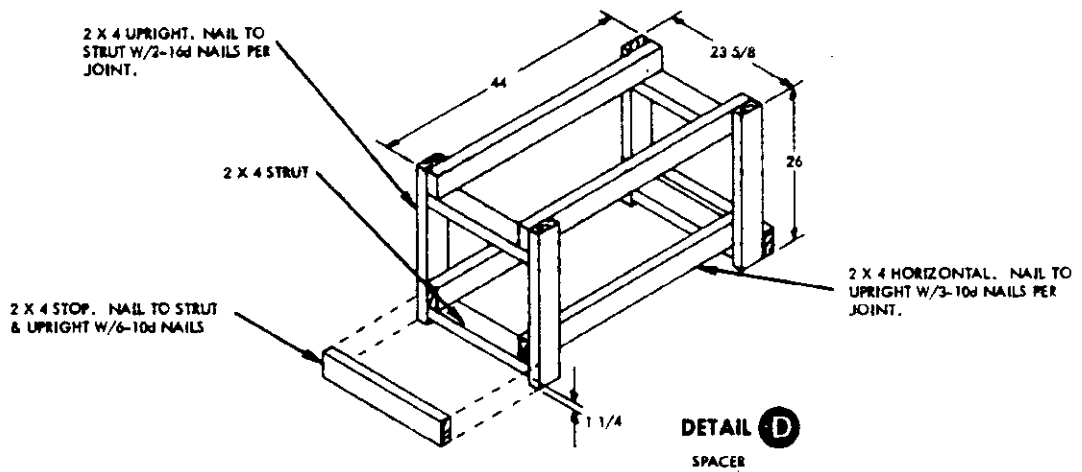
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UNITIZING



CONTAINERS STACKING AND BANDING PROCEDURE:

1. STACK TWO CONTAINERS USING A FORK LIFT TRUCK OR SUITABLE HOISTING DEVICE EQUIPPED WITH A SUITABLE FOUR LEG BRIDLE.
2. BE CERTAIN THAT THE NESTING FEATURES AND BEARING SURFACES ARE PROPERLY POSITIONED TO PREVENT SHIFTING IN LATERAL AND LONGITUDINAL DIRECTIONS.
3. SECURE BOTTOM CONTAINER TO TOP CONTAINER WITH TWO 1 1/4 X .035 STEEL STRAPS, POSITIONED AS INDICATED ABOVE.
4. AFTER TENSIONING STRAPS, SEAL WITH TWO 1 1/4" SEALS. DOUBLE CRIMP SEALS OR ONE DOUBLE NOTCHED SEAL.



DETAIL (D)

SPACER

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N199)

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